

Machine Learning

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Lecture 02: Supervised and Unsupervised Learning

Types of Machine Learning

- **Supervised Learning:** Learning from labeled data. The model is trained on input-output pairs.
- Requires labeled training data
- Goal: Map input to output
- Image classification (e.g., identifying cats vs. dogs)
- Spam detection in emails
- Regression tasks (predicting house prices)

Supervised Learning

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Heart disease
1	62	70	120	80	No
2	72	90	110	70	No
3	74	80	130	70	No
4	65	120	150	90	Yes
5	67	100	140	85	Yes
6	64	110	130	90	No
7	69	150	170	100	Yes
8	66	125	145	90	?
9	74	67	110	60	?

Types of Machine Learning

- **Unsupervised Learning:** Learning from unlabeled data. The model finds patterns and structures on its own.
- No labeled output
- Goal: Discover underlying structure
- Clustering (e.g., customer segmentation)
- Anomaly detection (e.g., fraud detection)

Unsupervised Learning

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia
1	62	70	120	80
2	72	90	110	70
3	74	80	130	70
4	65	120	150	90
5	67	100	140	85
6	64	110	130	90
7	69	150	170	100
8	66	125	145	90
9	74	67	110	60

Classification vs Regression

- Two main types of supervised learning problems.
- **Classification:** Task of predicting discrete labels.
 - Task of predicting discrete labels.
 - Output is a class label
 - Classifying emails as spam or not spam
 - Diagnosing diseases (e.g., healthy vs. diseased)
 - Image recognition (e.g., classifying images of vehicles)

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Heart disease
1	62	70	120	80	No
2	72	90	110	70	No
3	74	80	130	70	No
4	65	120	150	90	Yes
5	67	100	140	85	Yes
6	64	110	130	90	No
7	69	150	170	100	Yes
8	66	125	145	90	?
9	74	67	110	60	?

Classification vs Regression

- Two main types of supervised learning problems.
- **Regression:** Task of predicting continuous values.
 - Output is a real-valued number
 - Output is a class label
 - Predicting house prices based on features
 - Forecasting stock prices
 - Estimating temperature

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Cholesterol Level
1	62	70	120	80	150
2	72	90	110	70	160
3	74	80	130	70	130
4	65	120	150	90	200
5	67	100	140	85	190
6	64	110	130	90	130
7	69	150	170	100	250
8	66	125	145	90	?
9	74	67	110	60	?

Rule vs Learning

- We can write a complicated set of rules
 - Works well for a while
 - Cannot adapt well to new emails
 - Program could be reverse-engineered and circumvented
- Learn the mapping between an email and its label using past labelled data
 - Can be retrained on new emails
 - Not easy to reverse-engineer and circumvent in all cases
 - Easier to plug the leaks

Formalizing the setup

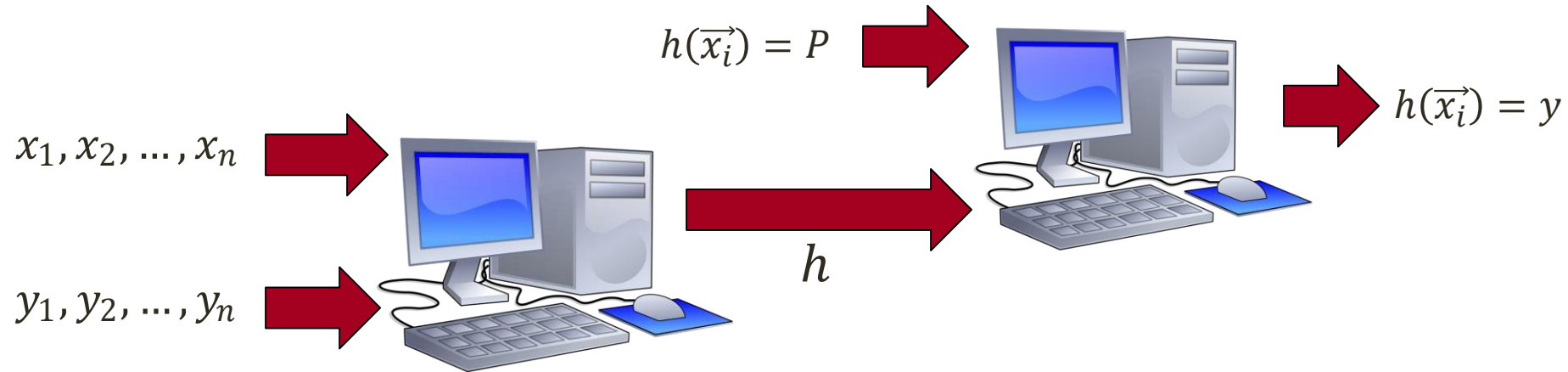
$$D = \{ x_1, y_1, x_2, y_2, \dots, x_n, y_n \} \subseteq X \times Y$$

- Where,
 - D is the data set
 - X is the d -dimensional feature space ($\mathbb{R}^d$)
 - x_i is the input vector of the i th sample
 - Y is the label space
- The data points are drawn from an unknown distribution P
$$(\vec{x}_i, y_i) \sim P(x, y)$$
- We want to learn a function $h \in H$, such that for a new instance $(\vec{x}_i, y_i) \sim P$
$$h(\vec{x}_i) = y$$
- With a high probability or at least, $h(\vec{x}_i) \sim y$

Training and Testing

Training

Testing



Feature Space

- **Student data (e.g. for predicting grades)**

$$\mathbf{x}_i = x_i^1, x_i^2, \dots, x_i^d$$

- Where each x_i^j is the value of the j th feature for patient i .
- Examples of features, x^j
 - Scores in assignments, quizzes, exams
 - Educational records, grades in previous courses
 - Rankings of previous educational institutes
 - Interaction with online tools? Missed instruments?
- Small number of features and relatively low features with 0 values:
Dense vectors

Feature Space

- Image data (e.g. for facial recognition)

$$\mathbf{x}_i = x_i^1, x_i^2, \dots, x_i^{3d}$$

- Where each group of three consecutive x_i^j s represent the red, green and blue values of each pixel in image i .
- In a 6-megapixel images, there are 6 million pixels and 18 million features
- Almost all pixels have non-zero values and the feature vector
- is mostly **dense**

x_i^1, x_i^2, x_i^3	x_i^4, x_i^5, x_i^6	x_i^7, x_i^8, x_i^9	...	
...	200, 89, 55	213, 22, 55	255, 7, 0	...
				$x_i^{58}, x_i^{59}, x_i^{60}$

Feature Space

- Textual data (e.g. for authorship attribution)

$$\mathbf{x}_i = \underset{i}{x^1}, \underset{i}{x^2}, \dots, \underset{i}{x^d}$$

- Where x^k or TF is the number of occurrences of the k th dictionary word in document i : Bag-of-Word features
- English has nearly 600,000 word-forms (273,000 headwords)
- Most of the feature values are zero: Sparse vectors

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.
<https://www.lipsum.com/>

a	0
abandon	0
ability	0
able	2
about	3
above	0
abroad	0
...	
Probably	0
product	1
production	0
profession	0
professional	0
professor	2
...	
youth	0
zest	0
zone	0
zoo	0

Feature One-hot vectors

- A single high value among zeros in the vector
- One-cold vectors (all 1's with one 0)

Binary	One-hot
000	00000001
001	00000010
010	00000100
011	00001000
100	00010000
101	00100000
110	01000000
111	10000000

Label Space

- **Binary (one-of-two) – Binary classification**
 - Sentiment: positive / negative
 - Email: Spam / Not Spam
 - Online Transactions: Fraudulent (Yes / No)
 - Tumor: Malignant / Benign
 - $y \in 0,1$ e.g. 0: Negative class, 1: Positive class
 - $y \in -1,1$ e.g. -1: Negative class, 1: Positive class
- **Multi-class (one-of-many, many-of-many problems) – Multi-class classification**
 - Sentiment: Positive / negative / neutral
 - Emotion: Happy, Sad, Surprised, Angry,...
 - Part-of-Speech tag: Noun / verb / adjective / adverb / ...
 - Recognize a word: One of $|V|$ tags
 - $y \in 0,1,2,3, \dots$ e.g. 0: Happy, 1: Sad, 2, Angry,...
- **Real-valued – Regression**
 - Temperature, height, age, length, weight, duration, price...

Hypothesis Space

- The hypothesis h is sampled from a hypothesis space H .
 - $h \in H$
- H can be thought of to contain classes of hypotheses which share sets of assumptions like
 - Decision tree
 - Perceptron
 - Neural networks
 - Support Vector Machines
- For example, $h \in H$ for H : Decision trees, would be instances of decisions trees of different height, arity, thresholds, etc.

So, how do we choose our h ?

- Randomly?
- Exhaustively?

How do we evaluate h ?

How to choose h ?

- Randomly
 - May not work well
 - Like using a random program to solve your sorting problem
 - May work if H is constrained enough
- Exhaustively
 - Would be very slow
 - The space H is usually very large (if not infinite)
- H is usually chosen by data scientists (you!) based on their experience!
 - $h \in H$ is estimated efficiently using various optimization techniques

Thank You